APID Weekly Tutorial Instructions

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Session 1: Opportunity Identification (Part 1)

1.1 In-class Activities

1.1.1 Activity 1

The objective of this activity is to walk you through the process of selecting a product for the mock development process as well as determine the company under which the product will be developed. At the end of the activity, your group should establish a clear direction for developing the chosen product within the specified company. Accuracy is not a priority at this stage, so you only need to draft in bullet point format.

1. During the discussion, decide on the product that your group intends to develop, consider the market demand, feasibility, and the group's capabilities in developing the product.

   You can select any product you want to design,

   (a) but it is recommended to select something the team members like and know well and have interest to develop.

   (b) The product can be an original (please, check Google)! OR a derivative of an existing product. But for simplicity, better find something that already available on the market

   (c) We can discuss the suitability of the product, but your team is responsible for selecting a product whose design is feasible given the constraints of the course and the requirements for this project.

   (d) While products can take various forms, this activity specifically emphasizes engineered, discrete, and physical products.

2. Determine the company under which this product will be developed. In this project, we assume each group affiliated with a firm and you should envision yourself as part of the chosen company.

   (a) For example, if your team chose to develop a smartphone (such as the Samsung Galaxy S series), you should imagine yourself as a Samsung employee.

   (b) The chosen firm can fall into one of two potential scenarios:

   - being a successful, well-established company that has developed the chosen product, or
   - being a small company aiming to enter the market for that product.

1.1.2 Activity 2

For those of you who are willing to paddle the extra mile, there are several sub-activities that can be conducted further. It should be noted that these activities are not required to be completed during class, and some of them may take several days to finish. The activities can help the team in selecting the product to be develop or may even serve as the foundation for your coursework project.

1. Draft the following which is related to customer needs identification:

   - Identify the specific needs that the product aims to address.
   - Identify how the product will provide a solution to the identified customer needs.

2. Discuss and outline the design of the product:

   - Specify the design requirements and parameters.

3. Estimate the number of parts required for the product.

   - create a rough sketch of the design, if possible.

4. Determine the sales price for the product:

   - Consider the manufacturing cost, desired profit margin, and market competition.

Session 2: Stage Gate

2.1 Pre-class Activities

2.1.1 Activity 1: Selecting product development process

Objective:
The objective of this activity is to help you in understanding different types of product development processes and subsequently adapt and tailor the development process according to your chosen project.

1. Study and become familiar with various types of product development processes, such as:
   - Technology-push products
   - Platform products
   - Process-intensive products.
   - Customized products
   - High-risk products
   - Quick-build products
   - Product-service systems
   - Complex systems
2. Determine which development process closely aligns with your team's chosen product.
   (a) You should consider the characteristics and requirements of your product and compare them with the information presented in Exhibit 2-4.
   (b) This analysis will help you identify the most suitable development process for your project.

2.1.2 Activity 2: Selecting Product Development Process Flows

Objective:
The objective of this activity is to guide you in understanding different types of Development Process Flows (DPS) and subsequently help in selecting DPS that most suitable one for your team's needs in developing the selected product. By understanding these process flows will enable you to effectively plan and execute the product development process.

1. Familiarize yourself with different types of Development Process Flows:
   - Generic process
   - Spiral product development process
   - Complex system development process
2. Analyze the characteristics and requirements of your team's selected product.
   (a) Consider factors such as complexity, risk, and flexibility in the development process.
3. Evaluate each Development Process Flow in relation to your product's needs and assess how each process flow aligns with the specific requirements and constraints of your team project.
4. Select the most suitable development process flow, for your team.

2.1.3 Activity 3: Selecting departmental structure

Objective:
The objective of this activity is to provide instructions on understanding the composition of a product development team and the departmental structure during the Rally Point phase. By familiarizing yourself with the interdisciplinary nature of product development and assessing the departmental needs for your project, you will be able to effectively organize and allocate resources within your team.

1. Familiarize yourself with the interdisciplinary nature of product development, which involve the contributions from various department within a company inclusive 1) marketing and sales, 2) engineering, 3) quality assurance, 4) manufacturing, 5) purchasing, 6) legal, 7) financial, and 8) project management.
   (a) You can refer to Exhibit 1-2 (Development of an electromechanical product) or Exhibit 2.7 (Tyco International) explore the different departmental structure during the rally point various types of product development projects.
2. For your own product development project, assess whether it is necessary to maintain, reduce, or add new departments to this list. Provide a well-reasoned explanation for your chosen approach.

2.1.4 Activity 4: Selecting Organizational structure

Objective:
The objective of this section in the lab manual is to guide you in understanding various organizational structures and selecting the most appropriate one for your specific product development project. By familiarizing yourself with these structures and making an informed decision, you will be able to effectively organize and manage your project team.

1. Begin by familiarizing yourself with different type of organizational structures which include 1) functional organization, 2) project organization, 3) lightweight organization, or 4) heavyweight project organization.
2. Determine the most appropriate organizational structure for your specific product development project. Take into account the choice you made in Activity 1 and Activity 2 from week 2.

2.2 In-class Activities

2.2.1 Activity 1: Selection of Product Development Strategy

Objective:
The objective of this activity is to walk your team with the process of reviewing and selecting the appropriate product development process in a group setting. Toward the end, it is expected the team will reach a consensus on the product development process.

1. At least, when joining this session, it is safe to say to each member of the team should already have generated multiple ideas related to the process of product development, the flow of the product development process, and the structure of different departments involved.
2. With this assumption, by considering ideas generated by each of the team member, discuss in group to determine
   (a) The appropriate product development process
   (b) The optimal flow of the product development process
   (c) The optimal flow of the product development process, and
   (d) The suitable departmental structure.
3. Do a proper justification for each of selection and compile the finalized selection and submit your group report on the ITEL Page.

Table 1: Types of Product Development Processes with Their Features and Examples

Process Type	**Description**	**Distinct Features**	**Examples**
Generic (Market-Pull) Products	The team begins with a market opportunity and selects appropriate technologies to meet customer needs.	Process generally includes distinct planning, concept development, system-level design, detail design, testing and refinement, and production ramp-up phases.	Sporting goods, furniture, tools.
Technology-Push Products	The team begins with a new technology, then finds an appropriate market.	Planning phase involves matching technology and market. Concept development assumes a given technology.	Gore-Tex rainwear, Tyvek envelopes.
Platform Products	The team assumes that the new product will be built around an established technological subsystem.	Concept development assumes a proven technology platform.	Consumer electronics, computers, printers.
Process-Intensive Products	Characteristics of the product are highly constrained by the production process.	Either an existing production process must be specified from the start, or both product and process must be developed together from the start.	Snack foods, breakfast cereals, chemicals, semiconductors.
Customized Products	New products are slight variations of existing configurations.	Similarity of projects allows for a streamlined and highly structured development process.	Motors, switches, batteries, containers.
High-Risk Products	Technical or market uncertainties create high risks of failure.	Risks are identified early and tracked throughout the process. Analysis and testing activities take place as early as possible.	Pharmaceuticals, space systems.
Quick-Build Products	Rapid modeling and prototyping enables many design-build-test cycles.	Detail design and testing phases are repeated a number of times until the product is completed or time/budget runs out.	Software, cellular phones.
Product-Service Systems	Products and their associated service elements are developed simultaneously.	Both physical and operational elements are developed, with particular attention to design of the customer experience and the process flow.	Restaurants, software applications, financial services.
Complex Systems	System must be decomposed into several subsystems and many components.	Subsystems and components are developed by many teams working in parallel, followed by system integration and validation.	Airplanes, jet engines, automobiles.

Session 3: Opportunity Identification

3.1 Pre-class Activities

As the name suggests, this is an activity that each member should do prior to the class meeting.

3.1.1 Activity 1: Generate Charter

Objective:
The objective of this activity is to guide you in creating and refining an innovation charter for your team project.

The innovation charter serves as a document that articulates the goals and boundary conditions for your innovation effort. For example, the charter for the FroliCat effort was:

> "Create a physical product in the cat toy category that we can launch to the market within about a year through our existing retail sales channel."
>
> –charter for the FroliCat

To begin the task,

1. Draft one or more innovation charters that you believe will be advantageous for your project.
   (a) Keep in mind that it is recommended to create a charter with a broad scope which will allow for a broad focus in the innovation charter and subsequently opens up opportunities that the team may not have considered initially.
   (b) Also, ensure the charter to be broader than what the team may initially feel comfortable with as this will challenge your team's assumptions about the types of opportunities that your team should pursue.
2. After drafting, it is advisable to review and refine them as necessary.
   (a) Pay attention to the clarity of the goals and the boundary conditions established in the charters.
3. Assume that there are several innovation chart, choose the most suitable innovation charter.
   (a) Consider the goals and boundary conditions that align best with your project's objectives and constraints.

3.2 In-class Activities

3.2.1 Activity 1: Generate Charter

During this stage, it is expected that each group member has individually generated several innovation charters according to the given guidelines.

1. Discuss with all members whether it is more advantageous to select the best innovation charter that had been drafted by one of the members, or to combine elements from multiple charters into a concise sentence.
   (a) Consider the diverse perspectives of each group member during the discussion.
   (b) Evaluate the benefits and drawbacks of choosing a single charter versus combining elements from multiple charters by take into account the shared vision, goals, and desired outcomes of the project.
2. Once consensus is reached, draft a new innovation charter that reflects the collective agreement of all team members.
   (a) Ensure that the new innovation charter is clear, concise, and representative of the team vision and objectives.
   (b) Review and refine the drafted innovation charter to ensure that it aligns with the desired outcomes and sets appropriate boundaries for the innovation effort.

Session 4: Opportunity Identification (Part 2)

4.1 Pre-class Activities

4.1.1 Activity 1: Generating List of Innovation Opportunities

Objective:
The objective of this activity is to guide you in creating opportunities for innovation based on the innovation charter you developed in Week 3. After completing this activity, it is expected that you can explore new ideas that align with the objectives and boundaries outlined in the charter.

To create opportunities for innovation based on the innovation charter you drafted in Week 3, follow these steps:

1. Refer to the innovation charter you developed in Week 3 as a guide for generating new opportunities.
2. Generate a minimum of three opportunities for innovation that align with the objectives and boundaries outlined in your innovation charter.
   (a) Try to think outside the box (without the help of any LLM) and explore multitude possibilities.
   (b) Consider various aspects including market needs, technological advancements, customer preferences, or process improvements.
   (c) Most important, ensure that the opportunities you come out should within the confine of the broad scope of the innovation charter and have the potential to bring value and impact to your team project.
3. Note-down each opportunity.

   While writing down each opportunity, consider the following optional steps, which are highly recommended to facilitate ease of revision.

   (a) Include a descriptive explanation of the idea and its relationship to the innovation charter.
   (b) Consider the potential benefits, challenges, or risks, and identify any necessary resources or support required to pursue each opportunity.

4.2 In-class Activities

4.2.1 Activity 1: Compiling List of Innovation Opportunities

During this stage, it is expected that each group member has individually generated several opportunities for innovation based on the individually drafted innovation charter.

1. Refer to the innovation charter that the team developed in Week 3 as a compass for this activity.
2. Review and evaluate the identified opportunities.
   (a) Keep in mind that the objective of this screening process is not to select a single "best" opportunity but to eliminate opportunities with low potential for value creation, while focusing opportunities that warrant deeper exploration.
   (b) Remember that the choices made during this screening step are not final and can be revisited and revised later in the process as the project progresses and more insights are gained.
   (c) A stickers will be provided to help the voting, but you may consider alternative voting strategies if desired.
3. Write down the finalized opportunities, along with their prioritization and rationale.

Session 5: Product Planning (Part 1)

5.1 Pre-class Activities

The pre-class activity requires you to engage in preliminary analysis and information gathering to prepare for product planning discussions.

5.1.1 Activity 1: Market Analysis and Planning Elements

Objective: The objective of this activity is to analyze key elements for product planning, including market segments, technology integration, manufacturing and service objectives, target market & price, and assumptions & constraints. This analysis will help in identifying promising opportunities and effectively managing your project portfolio.

To prepare for product planning, address the following questions:

1. **Market Segments:**
   (a) Identify and list potential market segments that should be considered when designing the product and its features.
2. **Technology Integration:**
   (a) Explore and identify new technologies, if any, that should be integrated into the new product.
3. **Manufacturing and Service:**
   (a) Define the manufacturing and service objectives and constraints that will impact product development.
4. **Target Market and Price:**
   (a) Clearly define the primary and secondary target market(s) for the product.
   (b) Establish a realistic target price range for the product in the defined market(s).
5. **Assumptions, Constraints, and Stakeholders:**
   (a) Outline key assumptions underlying your product plan.
   (b) Identify any constraints that may limit product development or market entry.
   (c) List the key stakeholders involved in the product planning process.

   Instructions:

   (a) Conduct independent research and brainstorming to address each of the points above for your chosen product or product idea.
   (b) Document your initial thoughts, findings, and preliminary answers for each question.
   (c) Be prepared to share and discuss your analysis with your team members during the in-class activity.

5.1.2 Activity 2: Mission Statement Draft

Objective: The objective of this activity is to draft a preliminary mission statement for your product, incorporating key details derived from your market analysis and planning considerations.

To draft a mission statement, include the following details:

1. **Product Description:**
   (a) Provide a concise description of the product itself.
2. **Benefit Proposition:**

(a) Clearly articulate the key benefit(s) the product offers to its target market(s).

3. **Key Business Goals:**

   (a) State the primary business goals the product is intended to achieve (e.g., market share, revenue, profitability).

4. **Primary Market:**

   (a) Identify the primary target market segment for the product.

5. **Secondary Markets:**

   (a) Identify any secondary market segments that the product may also address.

6. **Assumptions and Constraints:**

   (a) Briefly list the key assumptions and constraints identified in Activity 1 that are relevant to the mission statement.

7. **Stakeholders:**

   (a) List the primary stakeholders who are relevant to the product's success.

Instructions:

(a) Based on your analysis from Activity 1, draft a preliminary mission statement that incorporates all the details listed above.

(b) Prepare to share and refine your mission statement draft with your team members during the in-class activity.

5.2 In-class Activities

5.2.1 Activity 1: Collaborative Refinement of Market Analysis and Planning Elements

1. **Market Analysis and Planning Discussion:**

   (a) Share and discuss individual findings from Pre-class Activity 1 regarding market segments, technology integration, manufacturing and service objectives, target market price, and assumptions constraints.

   (b) Engage in a collaborative discussion to refine and consolidate the analysis based on the diverse perspectives and insights of team members.

   (c) Work towards reaching a consensus on the most relevant market segments, technologies, manufacturing considerations, target market price, and key assumptions constraints for your product plan.

5.2.2 Activity 2: Finalizing the Mission Statement through Team Collaboration

1. **Mission Statement Development and Finalization:**

   (a) Review and discuss the preliminary mission statement drafts prepared in Pre-class Activity 2 by each team member.

   (b) Through collaborative discussion, combine the best elements from individual drafts and refine the mission statement to create a unified and comprehensive version.

   (c) Ensure the final mission statement clearly articulates all required details: product description, benefit proposition, key business goals, primary market, secondary markets, assumptions and constraints, and stakeholders.

   (d) Confirm that the finalized mission statement reflects the collective agreement of all team members and provides a clear direction for further product planning activities.

5.3 Remark (Optional - For Extra Consideration or extra mark)

To demonstrate a deeper understanding and potentially earn extra mark be it for individual or group submission, you can add remarks that provide further justification, context, or insights to your analysis in either the pre-class or in-class activities.

Guidelines for Remarks:

1. **Detailed Justification:** If you make a specific choice or assumption in your analysis (e.g., selecting a particular target market, choosing a specific technology), provide a detailed justification for this decision. Explain why this is a suitable choice and support your reasoning.

2. **Reference/Citation:** Back up your justifications or insights with credible sources. This could include:

   - **Internet Sources**: Cite reputable websites, industry reports, market research data, or news articles relevant to your product or market. Ensure the sources are credible (e.g., Statista, Gartner, industry-specific publications, academic institutions).
   - **Journal Articles/Academic Research:** If applicable, cite relevant academic articles or research papers that support your analysis or provide theoretical frameworks. You can use databases like Google Scholar, JSTOR, or IEEE Xplore to find relevant publications.
   - **Textbook or Course Materials:** Refer to concepts, frameworks, or examples discussed in your textbook or other course materials to strengthen your arguments.

3. **Beyond the Basics:** Remarks should go beyond simply answering the basic questions. They should demonstrate critical thinking, industry awareness, and a deeper engagement with the product planning process. Examples of valuable remarks could include:

   - Discussing potential risks or challenges associated with your chosen market segment or technology.
   - Analyzing the competitive landscape and how it influences your product planning decisions.
   - Exploring relevant industry trends and how they might impact your product's success.
   - Justifying your chosen target price point based on market research or competitor analysis.
   - Explaining the rationale behind prioritizing certain stakeholders over others.

4. **Proper Citation Format:** Use a consistent citation format (i.e, APA) when referencing external sources. Even informal internet sources should be clearly linked or described.

Session 6: Product Planning (Part 2)

In class presentation.

Im here just to make sure the Session running number is tally as what I disclose in ITEL

Session 7: Customer Needs

Identifying customer needs is an integral part of the concept development phase of the product development process. The resulting customer needs are used to guide the team in establishing product specifications, generating product concepts, and selecting a product concept for further development.

Creating a high-quality information channel from customers to the product developers ensures that those who directly control the details of the product, including the product designers, fully understand the needs of the customer.

Given the quick timeline of our group assignment, it's essential to manage our expectations realistically. To fulfill the requirements for week 5's activity, you could, if feasible for your project, engage with potential customers to understand their needs. Alternatively, you may employ your creativity by envisioning yourself as the customer or utilize a Large Language Model (LLM) to generate possible customer needs.

Should you opt for the LLM approach, please ensure to document/record your prompts and incorporate these prompts into your final submission, both in the ITEL and the whiteboard presentation. This documentation will be invaluable for both myself and your classmates.

7.1 Pre-class Activities

7.1.1 Activity 1: Generating Question and Prompt for Survey Response

Objective:
The objective of this activity to generate useful questions and prompts for a survey response. These questions will help your team in collecting insights related to the usage of the selected product.
To begin

1. Consider the purpose and objectives of the survey. Such that, what specific information or insights you intend to obtain from the respondents. For example, it can be about typical use, likes, dislike, or suggested improvement

2. Based on the survey's goals, brainstorm potential questions and prompts that can help extract relevant and valuable responses from respondents.

3. For this activity, select a atleast three questions that you believe would be most appropriate for the survey response.

   (a) Keep in mind that the question should align with the objectives and the value the question would provide. And it should be about typical use, likes, dislike, or suggested improvement

   (b) It is beneficial if you can provide a justification explaining why you believe each of the question is appropriate for the survey response.

   (c) Ensure that the questions are clear, concise, and unbiased to obtain accurate and meaningful survey responses.

4. Note down the selected questions and their justifications.

Here is one example, the output generated from this activity

1. Question: How frequently do you use the product/service? Please select one option.

   - Daily
   - Weekly
   - Monthly
   - Occasionally
   - Never

2. Justification:
   This question helps us gather insights on the frequency of product/service usage. Understanding usage patterns allows us to identify customer engagement levels and potential areas for improvement. The response options are clear and structured, making it easy for respondents to convey their usage frequency accurately.

7.1.2 Activity 2: Needs Statement Generation

Objective:
The objective of this activity is to generate needs statements for the selected product.
To begin

1. Imagine performing an everyday task using the chosen product but with an assumption that you encounter frustrations and difficulties performing the task. Based on this imaginative experience, identify the needs that the product's developer may have overlooked. Based on this experience, generate a customer statement.
2. If time permits, observe someone else performing an everyday task using the chosen product and pay special attention to the frustrations and difficulties that person encounters.
   (a) Seek feedback from that person and generate a customer statement.
3. Once you have generated the customer statements either by imaginative experience or feedback from a person, translate these customer statements into proper needs statements:
   (a) Ensure that the needs statements are concise, specific, and clearly articulate the underlying requirement.
   (b) Use clear and unambiguous language to accurately represent the customer's needs.
   (c) Avoid any assumptions or biases when translating the customer statements into needs statements.

Customer Statement	**Need Statement**
"Why don't you put protective shields around the battery contacts?"	The screwdriver battery is protected from accidental shorting
"I drop my screwdriver all the."	The screwdriver operates normally after repeated dropping
"It doesn't matter if it's raining, I still need to work outside on Saturdays."	The screwdriver operates normally in the rain
"I'd like to charge my battery from my Power backup."	The screwdriver battery can be charged from a Power backup adapter
"I hate it when I don't know how much juice is left in the batteries of cordless tools"	The screwdriver provides an indication of the energy level of the battery

Table 2: Customer Statements and Corresponding Need Statements

7.1.3 Activity 3: Lead and Exterme User Consideration

Objective: Lead and Exterme User Consideration
The objective of this activity is reflect whether is essential to include lead users and extreme users in a survey related to your team's chosen product.
To begin

1. Reflect the objectives of the survey and the concept of lead users and extreme users.
2. Consider the following points to determine whether it is essential to interview these two user groups based on objectives of the survey:
   (a) Lead Users:
      - Assess whether lead users possess any attribute (e.g., unique insights, expertise, or experiences) that can contribute to understanding emerging needs and trends as well as the influence they may have on shaping future product developments or improvements.
   (b) Extreme Users:
      - Assess whether extreme users can provide valuable insights into edge cases or challenging scenarios that may not occur in regular usage as well as their ability to help identify areas of improvement or innovation that cater to specific user segments.
3. Note down your analysis and justification.

7.1.4 Activity 4: Exploring the Relationship between Identifying Customer Needs and Innovative Product Concepts (optional)

Can the process of identifying customer needs lead to the creation of innovative product concepts? In what ways? Could a structured process of identifying customer needs lead to a fundamentally new product concept like the Post-it Note?

7.2 In-class Activities

7.2.1 Activity 1: Creating A Customer Needs List and Generating Prompts For Survey Response

Objective:
The objective of this activity is to perform a comprehensive customer needs analysis within your assigned group.

To begin

1. Discuss with all members whether it is more advantageous to select the prompts for the survey response. that had been drafted by one of the members, or to combine elements from multiple prompt into a concise sentence.

   Once consensus is reached, draft a new customer's "prompts for the survey response" that reflects the collective agreement of all team members.

2. Next, in table with 3 columns, compile all the customer statement and the interpreted needs according to the theme (typical use, likes, dislike, or suggested improvement), as shown in Table 3.

Table 3: Customer Statements and Interpreted Needs for Thermostat Design

Question/Prompt	Customer Statement	Interpreted Need	Variable Type
Typical uses	I have to manually turn it on and off when it gets too hot or cold.	The thermostat maintains a comfortable temperature without requiring user action.	Functional
	Each time I want to change the temperature, I need to adjust both thermostats in the house.	Any user input need not be made in multiple locations. (I)	Functional
Likes—current model	I like that I can change the temperature if the setting is too high.	The temperature setting is easy to control manually.	Usability
	It didn't cost a fortune.	The thermostat is affordable to purchase.	Cost
Dislikes—current model	I'm too lazy to learn how to figure it out.	The thermostat requires little or no user instruction or learning.	Usability
	I sometimes forget to turn it off when we leave the house.	The thermostat saves energy when no one is home. (†)	Energy Saving
	Sometimes the buttons don't register a push and I have to press it repeatedly.	The thermostat definitively registers any user inputs.	Reliability
Suggested improvements	I would like my iPhone to adjust my thermostat.	The thermostat can be controlled remotely without requiring a special device.	Connectivity
	I like to shift quickly between different options like energy saving or ultra comfort.	The thermostat responds immediately to differences in occupant preferences.	Responsiveness

While optional, you may create a new column at the end of the table, to indicate what type of need the interpreted needs (direct, latent, constant, variable, general, niche) is being addressed.

3. The team may consider alternative groupings and labels for the needs statement. Also, it is possible to consider LLM to help you in group the needs statement.

Session 8: Product Specifications

In last session's tutorial, we translated customer statements into customer needs. In this session, we will continue working with these needs, aiming to transform them into measurable metrics that the Product Development (PD) team can utilize effectively.

8.1 In-class Activities

8.1.1 Activity 1: Create a needs and importance Table

First, please prepare a table with columns labelled Number, Sub-component, Needs, and Importance. Below is an example of how the table might look as shown in Table 4:

Table 4: Suspension system needs and importance

No	Sub-component	Need	Imp.
1	The suspension	Reduces vibration to the hands.	3
2	The suspension	Allows easy traversal of slow, difficult terrain.	2

You should list the sub-components that your group has identified from analyzing customer needs. Also, for each need you identify, assign an importance level on a scale from 1 to 5, where 1 means less important and 5 means very important.

8.1.2 Activity 2: Defining Metrics for Smooth Product Functionality

Now, let's find the appropriate metrics for each of the customer needs. Due to time constraints, we will use the LLM (language learning model) to assist us. Remember, this course is not focused on prompt engineering or LLM, but we are using it here to save time. You might consider using the following prompt: click me.

The LLM may produce different results depending on the trial and which service (ChatGPT, Gemini, Claude, etc) you used. It is important for you to assess the accuracy and appropriateness of these results. Then, map the output in the Table 5 according to your evaluation.

Table 5: Bicycle Suspension Needs, Metrics, Units, and Justification

No	Need	Metric	Units	Justification
1	Reduces vibration to the hands	1. Attenuation from dropout to handlebar at 10 Hz	dB	High-frequency vibrations are most noticeable at the hands; reduction improves rider comfort.
		2. Maximum value from the Monster	g	Captures peak impact events to evaluate suspension performance.
		3. Minimum descent time on test track	s	Demonstrates performance under real trail conditions.
2	Allows easy traversal of slow, difficult terrain	1. Spring preload	N	Adjustability of preload helps with traction and control on rough, slow trails.
3	Enables high-speed descents on bumpy trails	1. Attenuation from dropout to handlebar at 10 Hz	dB	Reducing vibration at high speeds helps maintain control and comfort.
		2. Maximum value from the Monster	g	Measures extreme shock loads during descents.
		3. Minimum descent time on test track	s	Indicates efficiency and capability over fast, bumpy terrain.
4	Allows sensitivity adjustment	1. Damping coefficient adjustment range	N·s/m	Provides tunability for rider preference and terrain type.
5	Preserves the steering characteristics of the bike	1. Maximum travel (26-in. wheel)	mm	Excessive suspension travel can affect geometry and handling.
6	Remains rigid during hard cornering	1. Spring preload	N	A stiffer spring resists compression and maintains geometry.
		2. Lateral stiffness at the tip	kN/m	Prevents lateral flexing which would impair handling under load.
7	Is lightweight	1. Total mass	kg	Lower mass improves bike efficiency and handling.
8	Provides stiff mounting points for the brakes	1. Lateral stiffness at brake pivots	kN/m	Ensures braking forces do not deform the frame and affect control.

Just to make life interesting, you could modify the prompt to generate 'Subjective Metrics for Subjective Needs'.

8.1.3 Activity 3: Defining Metrics for Smooth Product Functionality

As I often remind you, do not fully trust everything that the LLM produces. According to the textbook, we should follow certain guidelines when creating product specifications. Therefore, please add a new column to your table and check if any suggestions meet the following criteria:

1. Metrics that Reflect Customer Needs.
2. Dependent Metrics for Design Freedom
3. Practical Metrics for Product Development
4. Subjective Metrics for Subjective Needs
5. Metrics for Market Comparison

Evaluate whether the response from the LLM fully or partially meets any of these guidelines. In the new column, mark the numbers 1, 2, 3, 4, or 5 to indicate which guideline(s) each suggestion aligns with.

8.1.4 Activity 4: Translate into need-metric matrix (Optional)

This is an optional task that you can choose to do at home. Next, using Microsoft Excel, you will need to rearrange the table from Activity 3 into a new format called a 'need-metric matrix'. Your matrix should look similar to the example provided in Table 6

Table 6: Mapping of suspension system needs to associated measurable parameters. An `x` indicates that the corresponding parameter addresses the specified need.

	Needs	Attenuation @ 10 Hz (dB)	Max from Monster (g)	Spring preload (N)	Min descent time (s)	Max travel (mm)	Damping adj. (N·s/m)	Lat. stiffness tip (kN/m)	Lat. stiffness pivot (kN/m)	Total mass (kg)	Headset axis (mm)	Steerer length (mm)	Wheel sizes (list)
1	Reduces vibration to the hands	x	x		x								
2	Allows easy traversal of slow, difficult terrain			x									
3	Enables high-speed descents on bumpy trails	x	x		x								
4	Allows sensitivity adjustment						x						
5	Preserves steering characteristics of the bike							x					
6	Remains rigid during hard cornering			x				x					
7	Is lightweight									x			
8	Provides stiff mounting points for brakes								x				
9	Fits wide variety of bikes, wheels, tires										x	x	x
10	Is easy to install												
11	Works with fenders												
12	Instills pride												
13	Affordable for amateur enthusiast												
14	Not contaminated by water												
15	Not contaminated by grunge												

8.1.5 Activity 5: Collect Competitive Benchmarking Information

Next, we will assess the metric values used by various competing companies. You may create a table similar to the following Table 7 with columns representing each competitor's product. For this table, each row should correspond to a metric we identified earlier.

Table 7: Comparison of Suspension Metrics Across Bicycle Models

Metric No.	**Need Nos.**	**Metric**	**Imp.**	**Units**	**ST Tritrack**	**Maniray 2**	**Rox Talx Quadra**	**Rox Talx Ti 21**	**Tonka Pro**	**Gunhill Head Shox**
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	3	dB	8	15	10	15	9	13
2	2, 6	Spring preload	3	N	550	760	500	710	480	680

Next, we will Input the metric values for each competitor's product in the respective columns. In actual product development process, these values should be extracted from competitors' catalogs and supporting literature. However, due to time constraints, we will once again use the Language Learning Model (LLM) to help us define the appropriate specifications for each customer need. It's important to remember that this class does not focus on prompt engineering or the mechanics of LLMs; however, we utilize this tool to efficiently manage our time.

You might consider using the following prompt to engage the LLM: clickme. The LLM is expected to generate outputs in various formats depending on the trial and platform used. You must critically evaluate the accuracy and appropriateness of these outputs using your judgment, then document these metrics in the table as instructed.

8.1.6 Activity 6: Collect Competitive Benchmarking Information

For better visualization, we can represent these values using graphs. This helps us to clearly see the range of metric values across different competitive products.

You may consider the following prompt click me to generate a bar chart to display the values adopted by various competitors for a specific metric. Alternatively, you can generate a grouped bar chart click me to show the values adopted by various competitors across a range of metrics related to the product. If you are using the free version of ChatGPT, it might provide you with Python code instead of a direct graph. You can either:

1. Ask a friend who has taken a computer programming class to run the code and share the results with you.

2. Use codabrainy or (trinke), which does not require any installation, to run the code yourself.

This exercise is intended to help you understand how to visually compare different values used by competing products. Remember, the image created by the LLM might not always be accurate, so it is important to verify the output yourself. You can also use Microsoft Excel to create these graphs for more precise results.

Session 9: Functional Decomposition and Solutions

9.1 Pre-class Activities

9.1.1 Activity 1: Decompose the Problem

Objective:

The objective of this activity is to analyse the selected product's primary functions and decomposing the problem to identify its major components or subsystems. By documenting the chosen approach for problem decomposition and creating function diagrams, you will gain a clearer understanding of the product's functional elements and their relationships.

1. Analyze the product's primary functions and identify the major components or subsystems that contribute to its operation.
   (a) This step involves gaining a clear understanding of the overall function of the product.
2. Choose one of the following approaches for problem decomposition:
   (a) Functional Decomposition
   (b) Decomposition by Sequence of User Actions
   (c) Decomposition by Key Customer Needs
3. Document (justify the selection) the chosen approach for problem decomposition and apply it to the selected product.
   (a) Use appropriate diagrams, such as function diagrams or flowcharts, to visually represent the decomposition process and the relationships between different components or subfunctions.
   (b) Make sure to use VERB when expressing the functional element of the individual operation and operation. For example
      - Make Coffee
      - Extract Coffer
      - Keep Coffee
      - Warm Water
      - Move Water
      - Give Heat
4. Also, it is wise to have a write up about how the functional element works for ease of understanding
5. Note that at this stage the goal is to describe the functional elements of the product without implying a specific technological working principle for the product concept.
6. Save the output as a PDF when submitting to ITEL

9.1.2 Activity 2: Search and Find Solutions

Objective:
The objective of this activity is to walk you in executing comprehensive searches and gathering solutions for critical sub-problems related to the selected product and ultimately the finding may help you in generating concepts that ensure the product's smooth functionality.

Procedure:

1. To conduct an external search, you may consider sources such as academic publications, industry reports, technical journals, and relevant online platforms.
   (a) The combination between search engine and appropriate search keywords/phrases that are specific to each sub-problem may help in pinpointing/retrieving relevant information.
2. Jot down and compile at least five different solutions for each critical sub-problem obtained from the external/internal search.
   (a) For the benefit of other reader (your team member, and module convener), it is strongly recommended to details, such as the source, author, publication date, and a brief description of each solution.

9.2 In-class Activities

9.2.1 Activity 1: Decompose the Problem

Compile all findings and discuss with all members to determine the best approach to decompose the problem into its major components or subsystems. Each member should explain their selection to their team members. Once consensus is reached, choose one of the following approaches for problem decomposition:

1. Functional Decomposition
2. Decomposition by Sequence of User Actions
3. Decomposition by Key Customer Needs

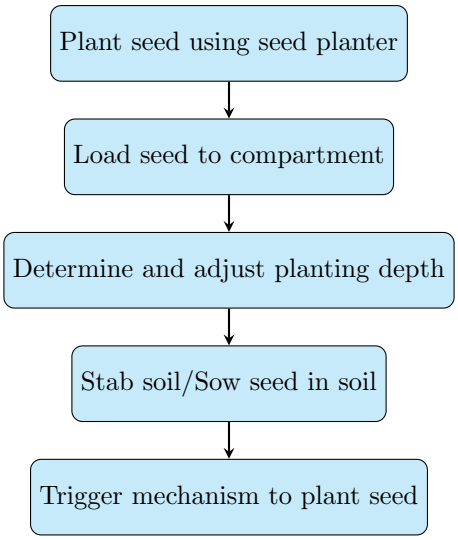

Figure 1: Minimum planting layout in Session 8

or another example of functional decomposition

code:importJson

```
Manual Seed Planter System
 Efficient Seed Storage and Dispensing Mechanism
    Seed Metering Plate
    Seed Metering Drum
    Auger-Based Funnel
    Vibratory Feeder
    Gravity Feed Chute
    Belt Conveyor with Seed Pockets (belt with holes carrying seeds)
    Air Suction Pickup (vacuum picks and releases single seeds)
    Peristaltic Tube Dispenser (seeds squeezed along flexible tubing)
    Cartridge Loading System (pre-loaded seed cartridges)

 Efficient Seed Sowing
    Syringe Tip Dispensing
    Spring Loaded Gate
    Flap Valve Mechanism
    Rotating Nozzle Dispenser
    Telescopic Tube Dispenser (tube that extends downward to sow)
    Flexible Nozzle Guide (rubbery nozzle that can be aimed precisely)
    Motorized Dropper (electric controlled seed dropping)
    Gravity Dropper Tube (simple tube relying on gravity but with guidance)

 Planting Depth Control
     Adjustable Depth Control Plate
     Depth Sensor Gauge
```

```
    Wheel-Based Depth Setter
    Adjustable Skid Shoe
    Spring-Loaded Plunger (presses the seed to a set depth)
    Soil Contact Sensor (detects soil pressure to adjust depth automatically)
    Telescopic Arm Adjustment (manual or motorized telescoping arm)
    Mechanical Stopper Rod (preset rod that limits downward travel)
```

9.2.2 Activity 2: Generate requirement vs function matrix.

Once a comprehensive list of project functions has been identified, the next crucial step is to assess how well these functions align with customer needs. It is equally important to ensure that the number of functions is balanced — sufficient to cover all requirements without introducing unnecessary complexity.

To systematically perform this evaluation, we develop a Requirement vs Function Matrix as shown in Table 8. This matrix provides a clear and organized method for mapping each customer requirement to the corresponding functions intended to fulfill them. It serves as a visual "roadmap," illustrating the connection between what needs to be achieved and how it will be accomplished.

Table 8: Requirements-Functions Matrix: Manual Seed Planter System

Requirements	Seed Metering Plate	Vibratory Feeder	Air Suction Pickup	Gravity Feed Chute	Telescopic Tube Dispenser	Motorized Dropper	Adjustable Depth Control Plate	Soil Contact Sensor
Precise dispensing of single seeds per planting event	X		X		X	X		
Minimal damage to seeds during dispensing process		X	X	X	X	X		
Accurate planting depth control for various seed types					X		X	X
Suitable for different seed sizes	X	X	X	X				
Reliable operation under different soil conditions					X	X	X	X
Easy manual adjustment for seed depth settings					X		X	
High throughput planting efficiency	X	X	X	X	X	X		
Minimal operator training and effort	X			X	X	X	X	
Robustness and minimal maintenance requirements	X	X		X			X	X

9.2.3 Activity 3: Search and Find Solutions

Compile all findings and discuss with all members to determine whether it is more advantageous to select the best different solutions for each critical sub-problem obtained from the external/internal search drafted by one of the members, or to combine findings from all team members. Once consensus is reached, compile at least five different solutions for each critical sub-problem obtained from the external/internal search that reflect the collective agreement of all team members.

Session 10: Concept Generation

10.1 Pre-class Activities

10.1.1 Activity 1: Create Concept Classification Trees OR Concept Combination Tables

Objective:

The objective of this activity is to walk you in developing either concept classification trees and concept combination tables. These tools help to better organize and categorize the identified solution concepts for each critical sub-problem, and also help to explore different combinations of sub-solutions.

Procedure:

1. Start this activity by first familiarizing yourself regarding the purpose and benefits of concept classification trees and concept combination tables.
2. For each critical sub-problem identified in activity 1, develop concept classification trees
   (a) Label the nodes or branches to indicate the different categories or attributes used for classification. Example of categories or attributes including functionality, cost, feasibility, user experience
   (b) Assign each solution concept to the appropriate category or attribute within the concept classification tree.
3. Develop concept combination tables to systematically explore different combinations of sub-solutions.
   (a) Create a table with
      - columns representing the sub-problems
      - rows representing the different solution concepts.
   (b) Identify and develop possible combinations by cross-referencing the sub-problems and solution concepts.

10.1.2 Activity 2: Generate rough product concepts

Objective:
The objective of this activity is to generate rough product concepts by combining the solution concepts from different sub-problems within the concept combination table

Procedure:

1. Refer to the output from activity 1, and generate rough product concepts by combining the solution concepts from different sub-problems within the concept combination table
   (a) Be creative and explore various combinations to generate a range of potential product concepts.
   (b) Provide brief description to effectively convey the key aspects of the proposed rough product concepts.
   (c) Provide information such as the technology utilized, working principles, intended functionality, and unique features of the concept.
2. Repeat steps 1 for at least five generated concepts that show promise or align well with the project goals for further development.

10.2 In-class Activities

10.2.1 Activity 1: Create Concept combination table

Compile all findings and discuss with all members whether it is more advantageous to select the best concept combination table that has been drafted by one of the members, or to combine elements from multiple concept combination tables to create a new concept combination table that systematically explores different combinations of sub-solutions. Once consensus is reached, draft a concept combination table that reflects the collective agreement of all team members.

10.2.2 Activity 2: Generate rough product concepts

Compile all findings and discuss with all members about the generated concepts that have been drafted by each one of the members. Select the top 5 generated concepts that show promise or align well with the project goals for further development.

Session 11: Concept Selection

11.1 Pre-class Activities

11.1.1 Activity 1: Generating selection Criteria

Objective:
The objective of this activity is to walk you through in establishing a set of metrics that can be used to narrow down the number of concepts while improving the overall quality. At the end of this activity, you should be able to generate criteria that can accurately differentiate between different concepts.

1. Review the chosen product:
   (a) By this time, it is expected that you have already familiarized yourself with the chosen product and its specifications.
2. Take note of any specific customer needs which are relevant to the product.
   (a) Refer to the customer needs (Session 5 activity) that have been identified for the chosen product.
3. Take note of any specific enterprise needs which are relevant to the product.
   (a) Refer to the enterprise needs that have been identified for the chosen product, if available.
4. Draft the selection criteria.
   (a) Choose at least one dimension that will serve as selection criteria for the concepts.
   (b) The selection criteria should effectively distinguish among the concepts.
   (c) Expressed the criteria at a high level of abstraction.
   (d) Do not include unimportant criteria in the screening matrix to avoid difficulty in interpreting results.

11.1.2 Activity 2: Creating a Concept-Screening Matrix

Objective: The objective of this activity to walk you through the process of setting up a concept-screening matrix which is critical when evaluating and comparing different concepts based on specific dimensions.
Lets start:

1. Gather the identified concepts:
   (a) Referring to the concepts that were identified during the Session 7 activity.
2. Determine the dimensions for the concept-screening matrix:
   (a) Refer to criteria selected on activity 1.
3. Set up the concept-screening matrix:
   (a) Take a blank A4 paper and orient it horizontally (landscape).
   (b) Draw a table with
      - columns for the identified concepts and
      - rows for the selected dimensions.
   (c) Label the
      - first column as "Concepts" and write the names or labels of the identified concepts in the subsequent rows.
      - the first row with the selected dimensions.

11.2 In-class Activities

11.2.1 Activity 1: concept-screening matrix

Objective: The objective of this activity to walk you through the process of reviewing and consolidating draft matrices in a group setting. By following these instructions, you will work towards reaching a consensus on the concept-screening matrix and proceed to finalize the selection matrix.

Procedure:

1. Gather the draft matrices prepared by each team member.
2. Assess the feasibility of merging the draft matrices or selecting the most promising elements from each.
3. Collaborate with the group to reach a consensus on the concept-screening matrix.
4. Once a consensus is reached, proceed to finalize the selection matrix.

11.2.2 Activity 2: Deciding on the Need for Concept Scoring

Objective:

The objective of this activity to walk you through the process of evaluating the need for concept scoring and making an informed decision as a group. By following these instructions, you will assess whether concept scoring is necessary for your project and understand the justification for not employing concept scoring if applicable.

If your team thinks concept scoring is necessary, they can find the steps for performing it in Chapter 8 under the section "Concept Scoring."

Procedure:

1. As a group, discuss the need for concept scoring.
   (a) Make sure every team member has a clear understanding of the purpose and benefits of concept scoring.
2. Evaluate whether increased resolution is necessary to better distinguish between competing concepts?
3. Justification for not employing concept scoring (if applicable):
   (a) Provide a statement to justify the decision should the group decides that concept scoring is not necessary.
   (b) The reason for can due to the level of differentiation already achieved, time constraints, available resources, or other valid reasons.
4. Process flow for concept scoring (if applicable):
   (a) Refer to the instructions and guidelines outlined in Chapter 8 of our textbook, to conduct concept scoring.

Session 12: Concept Testing

12.1 Pre-class Activities

12.1.1 Activity 1: Define the Purpose of the Concept Test

Objective: The objective of this is activity is to better understand the importance of generating questions that you aim to address during the concept testing phase. These questions should provide valuable insights and aid in decision-making

1. Draft the questions that your team aim to answer through the concept test

   The typical questions addressed in concept testing are:

   - Which of several alternative concepts should we pursue?
   - How can we improve the concept to better meet customer needs?
   - Approximately how many units are likely to be sold?
   - Should we continue with the development of the concept?

2. Document the question(s)

12.1.2 Activity 2: Choose a Survey Population

Objective: The objective of this activity is to identify the market segment or target audience for your product and select a survey population that closely represents the characteristics of your target market.

Procedure:

1. Search for relevant resource to help you understand the specific group of customers (market segment) who are most likely to be interested or benefit from your product.

2. Once gain an idea, proceed by selecting survey population that closely resembles the characteristics of your target market. This population should reflect your potential customers (e.g., age, gender, geographic location, preferences).

3. Determine the sample size for your survey population.

   (a) You use free online service such as click me to calculate the appropriate sample size for your research

   (b) While it is not necessary to survey everyone, it is imperative to take into account this statistical consideration.

12.1.3 Activity 3: Choose a Survey Format

Objective: The objective of this activity is to select a survey format and develop survey questions that are clear and relevant to the purpose of the concept test.

1. Consider the different survey formats available and choose the one that suits your needs. Some common survey formats include:

   - Face-to-face interaction
   - Telephone interviews
   - Postal mail surveys
   - Electronic mail surveys
   - Internet-based surveys

2. Draft the survey and possible interview questions that will be used during the survey

   (a) Keep in mind to ensure the questions are clear and relevant to the purpose of the concept test.

12.1.4 Activity 4: Communicate the Concept

Objective: To objective of this activity to decide on a communication method for presenting the concept to survey participants and ensure that the chosen method allows respondents to understand and evaluate the concept accurately.

1. Decide how you will communicate the concept to the survey participants by considering factor such as
   - nature of your product
   - the survey format you have chosen.
2. Try to come out with one or more means of communication that are suitable for presenting the concept effectively, options including
   - Verbal description
   - Sketches
   - Photos and renderings
   - Storyboards
   - Videos
   - Simulations
   - Interactive multimedia
   - Physical appearance models
3. Keep in mind that the choice of survey format is closely linked to how you communicate the product concept.
   (a) Make sure the chosen format allow respondents to understand and evaluate the concept accurately.
   (b) Select communication methods that effectively convey the key aspects and attributes of the concept and subsequently facilitating meaningful feedback from participants.

12.2 In-class Activities

12.2.1 Activity 1: Defining the Purpose of the Concept Test

1. Compile all the findings.
   (a) Each group member should already individually generate several question(s).
2. Discuss with all team members the most advantageous approach:
   (a) Decide whether to select the best question drafted by one member or combine elements from multiple questions to create a new single question or several questions.
3. Once a consensus is reached, draft a question(s) that reflects the collective agreement of the team.

12.2.2 Activity 2: Choosing a Survey Population

1. Compile all the findings.
   (a) Each group member should already individually generate several market segment and survey population.
2. Discuss with all team members whether it is more advantageous to
   (a) select the best market segment and survey population drafted by one member or
   (b) combine elements from market segments and survey populations drafted by all team members.
3. Once a consensus is reached, draft a market segment and survey population that reflects the collective agreement of the team.

12.2.3 Activity 3: Choosing a Survey Format

1. Compile all the findings.
   (a) Each group member should already individually draft their justification about the best survey format for the chosen product.
2. Discuss the advantage and disadvantage the proposal given by each member.
3. Once a consensus is reached, pick and documented the suitable survey format.

12.2.4 Activity 4: Communicating the Concept

1. Discuss with all team members whether it is more advantageous to select the best communication method proposed by one member or combine elements from multiple communication methods.
2. Once a consensus is reached, propose a communication method that reflects the collective agreement of the team.